

Ariella Vu

Senior Software Engineer

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PROFILE

Senior front-end and full-stack engineer with 11+ years of experience and hands-on application security. Security liaison for MetaMask's browser extension, serving 30M+ monthly active users, where I led the Manifest V3 migration ahead of Chrome's deprecation deadline and ran investigations into high-impact vulnerabilities. I set technical direction through written proposals, contribute upstream to open-source standards, and mentor the engineers around me.

EXPERIENCE

Senior Software Engineer II — MetaMask, Consensys

Remote | Open-source self-custody wallet, 30M+ monthly active users | January 2022 – September 2025

- **Manifest V3 migration:** Led the cross-browser migration, re-architecting the extension around service workers and Node.js multiplex streams to keep the wallet compliant and shippable ahead of Chrome's MV2 deprecation
- **Security liaison:** Served as security liaison for the extension, triaging incoming vulnerability reports, setting severity and response, and leading investigations into XSS, cross-domain phishing, Unicode-based data obfuscation, and JavaScript supply-chain compromise
- **Ethereum standards:** Contributed to Ethereum Improvement Proposals through reviews and upstream pull requests, generated smart contracts for JSON-RPC testing, and discussed technical specs and concerns with proposal authors
- **Release quality:** Caught correctness and quality defects in an already-approved change before release, then scoped the remediation epic and drove the rework through to ship
- **Transaction security:** Set the architecture for secure transaction-data UI in React, aligned with protocol specifications, so users can verify exactly what they are signing, and deployed transaction-simulation safeguards that prevent phishing attacks and unauthorized fund transfers
- **Financial accuracy:** Fixed a 7-year-old BigNumber precision defect and related precision bugs affecting the accuracy of the financial values users relied on
- **Confirmations redesign:** Piloted a redesign spanning 30+ modal views and shipped it in two weeks, clearing a backlog of long-standing UI defects in the same effort
- **Observability:** Found and fixed critical Mixpanel and Sentry misconfigurations affecting security, privacy, and data integrity across the product
- **Community and ecosystem:** Presented technical workshops, mentored at three external hackathon events involving hundreds of developers, and built external partnerships driving third-party Snaps integrations

Senior Software Engineer — Copper CRM

San Francisco, CA & Remote | B2B SaaS | August 2016 – April 2021

- **Technical direction:** Set client-side technical direction through 7 written architectural proposals, all later implemented, including complex refactors
- **Performance:** Cut rendering latency for datasets exceeding 10,000 records from 1250 ms to 41 ms (~97%), then presented the memory-leak and performance findings org-wide
- **A/B testing platform:** Served as client-side technical lead for a custom A/B testing platform, unlocking data-driven decisions across analytics, research, and feature flagging
- **Google partnership:** Acted as tech lead on the Google partnership, launching a Workspace add-on through their Beta Program and delivering a Chrome extension that put CRM records directly in Gmail
- **Product UI:** Led Ember.js development of accessible, responsive UI/UX including dashboards, infinite-scroll lists, first-time user experiences, and an in-app RingCentral VoIP service
- **Pricing tooling:** Built the responsive pricing page and packaging configuration tool, letting stakeholders change plans and pricing through Ruby on Rails behind feature flags

- **Hiring:** Interviewed 15+ candidates, authored the technical coding challenge, and refined the evaluation criteria the team hired against
- **Integrations:** Implemented third-party platforms including MixPanel, Pendo, Segment, and Google Tag Manager
- **AMP email:** Shipped AMP emails enabling recipients to reply or react with emojis from within their own inbox
- **Remote Work:** Pioneered remote work as the company's first remote employee in 2018

Software Engineer — Wayvia (formerly PriceSpider)

Irvine, CA | Retail commerce data platform | February 2015 – June 2016

- **Analytics application:** Launched a MEAN-stack AngularJS analytics application with RESTful APIs in 4.5 months, and designed the foundational PostgreSQL and MongoDB schemas
- **Brand widgets:** Developed 25+ responsive, cross-browser “Where to Buy” widgets used by global brands including Samsung, Canon, and Mars
- **Authentication:** Architected the authentication system and login flow with brute-force prevention via server-side session tracking and attempt limiting, plus role-based access control with granular read/write permissions
- **Data pipelines:** Authored resilient data-extraction services parsing store and product data from retailer sites across multiple languages, using request rotation and rate-limit handling to keep collection stable
- **Leadership:** Identified knowledge gaps across the engineering team, then created and hosted weekly AngularJS meetings to share performance techniques and best practices

SELECTED WORK

- **Manifest V3 migration:** Service-worker re-architecture of the MetaMask extension
github.com/MetaMask/metamask-extension/pull/15494, [pull/16075](https://github.com/MetaMask/metamask-extension/pull/16075)
- **BigNumber precision fix:** 7-year-old defect affecting financial accuracy
github.com/MetaMask/metamask-extension/pull/25741
- **Release remediation:** Defects caught in an approved change, scoped epic, shipped rework
[pull/14438](https://github.com/MetaMask/metamask-extension/pull/14438), [issues/18471](https://github.com/MetaMask/metamask-extension/issues/18471)

SKILLS

Languages: TypeScript, JavaScript, Python, HTML, CSS, Sass

Front-End: React, Redux, React Native, Next.js, Ember.js, AngularJS, Tailwind CSS, Storybook

Back-End & Data: Node.js, Express, GraphQL, REST API design, PostgreSQL, MongoDB, Ruby on Rails

Security: Application security, vulnerability research, XSS, phishing, JS supply-chain security, browser security, secure UI/UX, authentication & authorization

Tooling: Git, CI/CD, Jest, Sentry, Chrome DevTools, browser extensions (Manifest V3), Vercel, AI-assisted Development (Claude Code, Cursor, Sourcegraph)

Practices: System design, release management, performance optimization, accessibility, A/B testing, internationalization, cross-browser compatibility, Agile

EDUCATION

B.A.Sc., Computer Science — University of California, Davis | September 2010 – September 2014